

Revision Notes for Class 12 Chemistry

Chapter 7 – Alcohol Phenol and Ether

1. Introduction

Alcohols are compounds that have a hydroxyl group ($-OH$) attached to a saturated carbon atom. Enols are compounds that have a hydroxyl group attached to an unsaturated carbon atom of a double bond. The saturated carbon can be alkyl, alkenyl, alkynyl, cycloalkyl, or benzyl. If, on the other hand, a hydroxyl group is attached to a benzene ring. Phenols are the name given to these compounds.

The alcohols are further classified as monohydric (containing one $-OH$ group), dihydric (containing two $-OH$ groups), and trihydric (containing three $-OH$ groups) (containing three $-OH$ groups).

Alcohol is used in both industry and everyday life. Chiefly ethanol, for example, is a common spirit used to polish wooden furniture. Sugar, cotton, and paper are all composed of compounds that contain groups. Phenols are found in a variety of important polymers, including Bakelite, as well as pharmaceuticals such as Aspirin. Ethers are commonly used as anaesthetics and solvents.

In alcohols, the oxygen of the group is attached to carbon by a sigma (σ) bond formed by the overlap of a sp hybridised orbital of carbon with a sp hybridised orbital of oxygen. The following figure depicts structural aspects of methanol, phenol and methoxymethane.

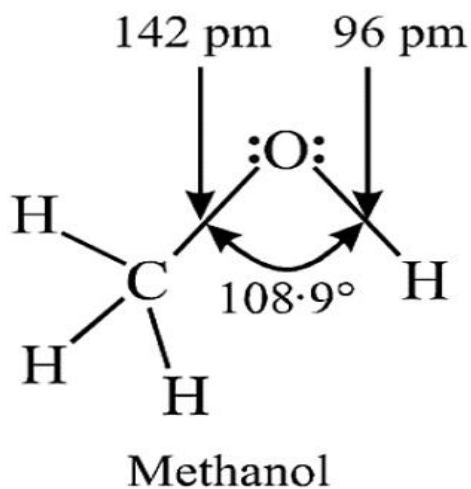
2. Classification

$\text{CH}_3\text{—OH}$	Methanol	
$\text{R—CH}_2\text{—OH}$	1° Alcohol	Primary
$\begin{array}{c} \text{R} \\ \\ \text{R—CH—OH} \end{array}$	2° Alcohol	Secondary
$\begin{array}{c} \text{R} \\ \\ \text{R—C—OH} \\ \\ \text{R} \end{array}$	3° Alcohol	Tertiary
R—O—R	Symmetrical Ether	
R—O—R'	Unsymmetrical Ether	

Classification of alcohols and symmetrical and unsymmetrical ether

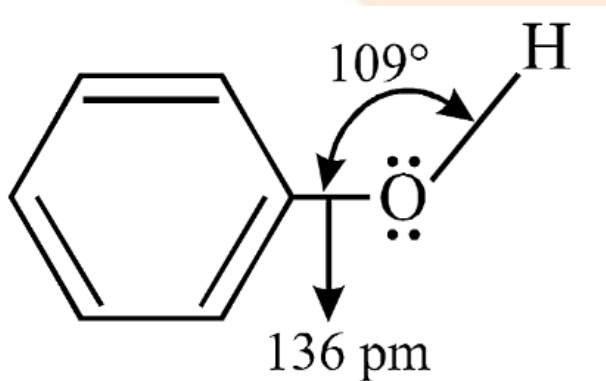
3. Structure of Functional Groups

In the structure of alcohols, the oxygen of hydroxyl group is connected to carbon through a sigma bond which is formed by the overlapping of sp^3 hybridised orbital of C with a sp^3 hybridised orbital of oxygen. The structural aspects of phenol, methanol and methoxymethane are depicted by the following figure.



Structure of methanol

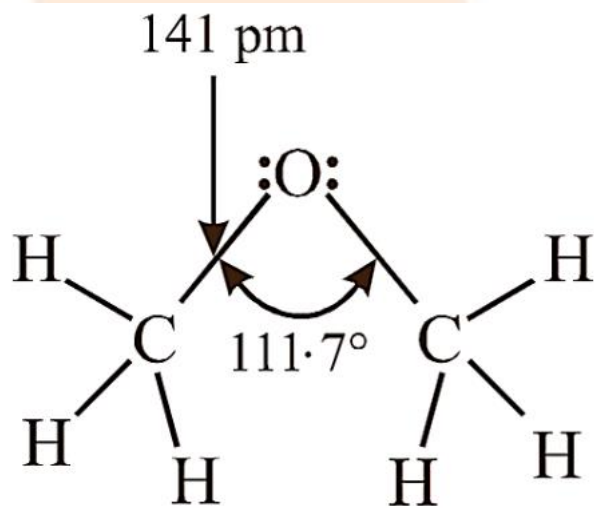
Due to lone pair-lone pair repulsion, bond angle is slightly less.



Phenol

Structure of Phenol

Since a pair of oxygen is delocalised on the ring, the length of the C - O bond is reduced.



Methoxymethane

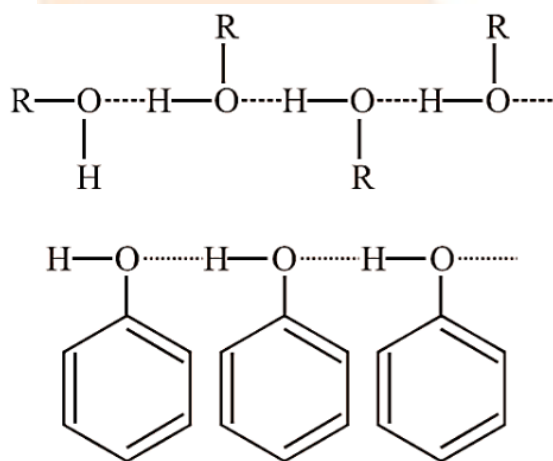
Structure of Methoxymethane

Due to the general repulsive interaction between the two bulky (R) groups, the bond angle in methoxymethane is greater than the tetrahedral angle. The length of the C - O bond is the same as in alcohols.

4. Physical Properties

4.1 Boiling Point

The boiling points of alcohols and phenols rise as the number of carbon atoms increases (increase in van der Waals forces). The boiling point of alcohol decreases as branching increases (decrease in Van der Waals forces due to decrease in surface area). In alcohols and phenols, the -OH group contains a hydrogen atom that is bonded to an electronegative oxygen atom. As a result, it is capable of forming intermolecular hydrogen bonds with greater strength than amine.



Formation of intermolecular hydrogen bonds in alcohols and phenols

Alcohols and phenols have higher boiling points than other classes of compounds, such as hydrocarbons, ethers, and haloalkanes/haloarenes, amines with comparable molecular masses, due to the presence of strong intermolecular hydrogen bonding.

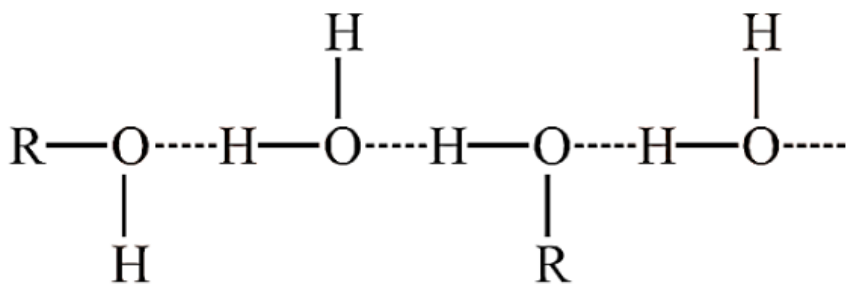
Their boiling points are lower than those of carboxylic acid, which has a stronger hydrogen bond. Boiling points for isomeric alcohols decrease as branching increases due to a decrease in van der Waals forces as size decreases. The boiling point sequence is primary alcohol > secondary alcohol > tertiary alcohol.

Due to lower dipole moment and the absence of H-bonding, the boiling point of ethers is very low and comparable to that of alkanes of comparable molecular mass.

4.2 Solubility

Alcohols and phenols are soluble in water due to their ability to form hydrogen bonds with water molecules. Solubility decreases as the size of the hydrophobic group increases (R). Higher concentrations of alcohol are insoluble. Because of the decrease in surface area of the nonpolar hydrocarbon part, branching increases solubility.

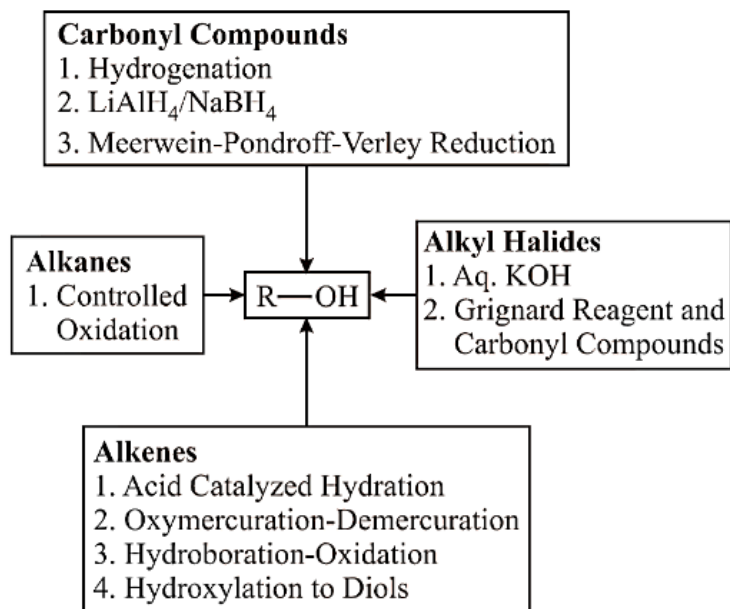
n-butyl alcohol < isobutyl alcohol < sec-butyl alcohol < tert-butyl alcohol



Formation of hydrogen bonds Between alcohol and water

Lower ethers are water soluble, but their solubility is lower than that of alcohol due to less H-bonding with water and being less polar.

5. Preparation of Alcohols



Flow chart showing preparation of alcohols

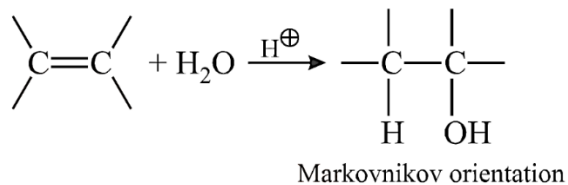
5.1 Alkane

Controlled oxidation

5.2 Alkenes

5.2.1 Acid Catalyzed Hydration

Markovnikov addition with carbocation rearrangements.

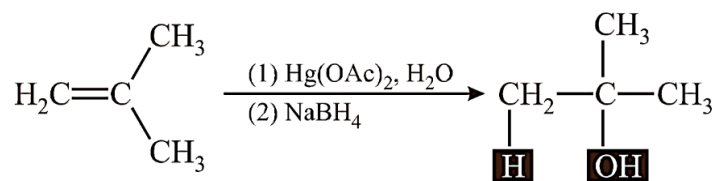


Reaction showing Acid Catalyzed Hydration of alkene

5.2.2 Oxymercuration-Demercuration

Markovnikov addition without carbocation rearrangements.

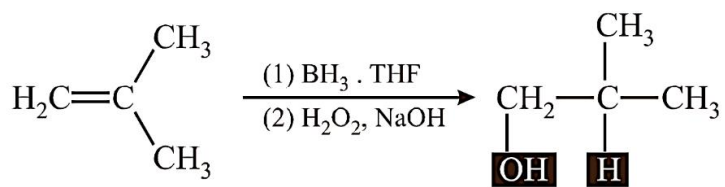
Example - 1



Reaction showing Oxymercuration-Demercuration

5.2.3 Hydroboration-Oxidation Anti-Markovnikov Addition.

Example-2

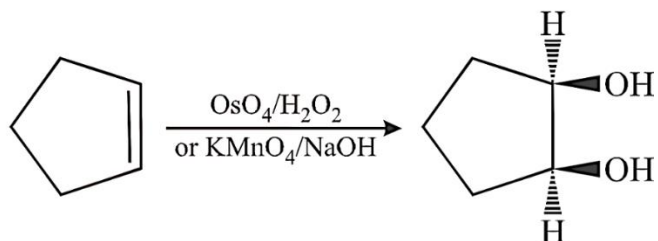


Reaction showing Hydroboration-Oxidation Anti-Markovnikov addition

5.2.4 SYN Hydroxylation

Reagents: Cold dil. $\text{KMnO}_4 / \text{NaOH}$ or $\text{OsO}_4 / \text{H}_2\text{O}_2$

Example-3

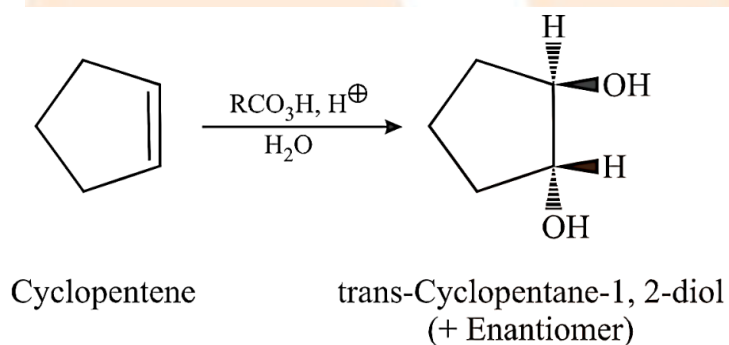


Reaction showing Syn Hydroxylation

5.2.5 ANTI Hydroxylation

Reagents: Peroxy acids followed by acidic hydrolysis

Example-4

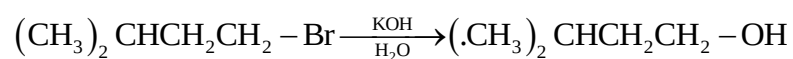


Reaction showing Anti Hydroxylation

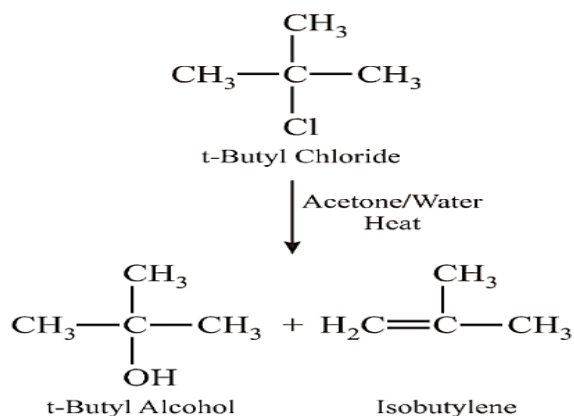
5.3 Alkyl Halide

5.3.1 Second Order Substitution

Primary and some Secondary Halides



Example -6



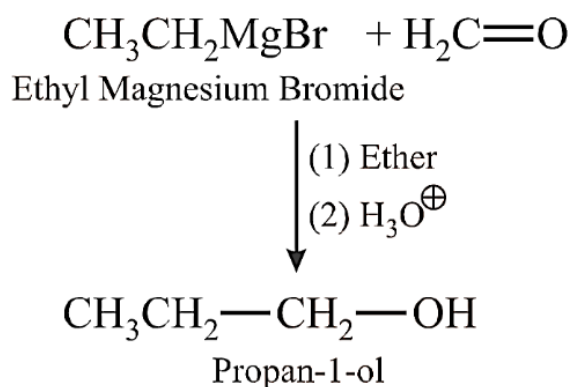
Reaction between tertiary butyl chloride and acetone

5.3.2 Grignard Reagent/ Organolithium Reagent

Nucleophilic addition to the carbonyl group

- a. Addition to Formaldehyde-primary alcohol

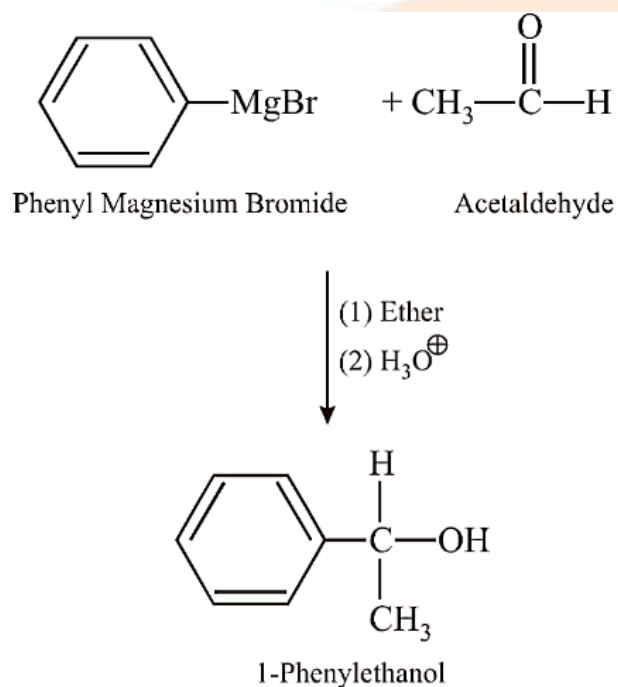
Example-7



Reaction showing addition of Grignard Reagent to Formaldehyde forming primary alcohol

b. Addition to aldehyde-secondary alcohol

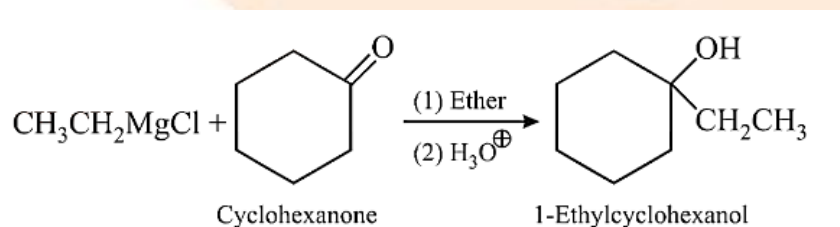
Example-8



Reaction showing addition of Grignard Reagent to aldehyde forming a secondary alcohol

c. Addition to a ketone-tertiary alcohol

Example-9

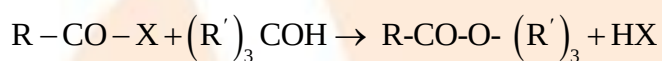


Reaction showing addition of Grignard Reagent to a ketone forming a tertiary alcohol

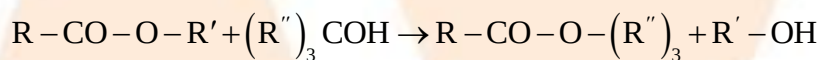
d. Addition to an acid-halide or an Ester-tertiary alcohol

Example-10

Addition to an Acid Halide



Addition to an Ester



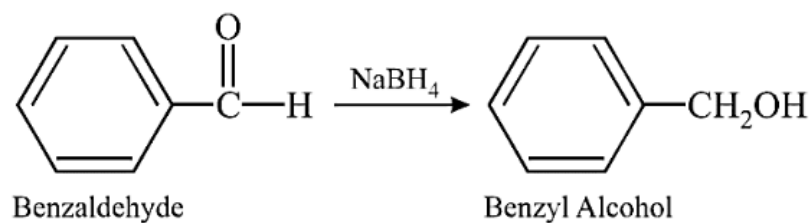
e. Addition to Ethylene oxide-primary alcohol (with two carbon atoms added)

Example-11



5.4 Carbonyl Compounds

5.4.1 Catalytic Hydrogenation



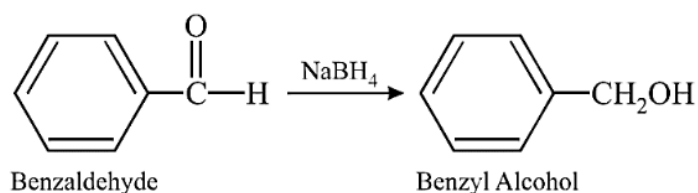
Reaction showing Catalytic Hydrogenation

This method is usually not known as effective or selective as the use of hydride reagents.

5.4.2 Reduction with Metal Hydrides

1. Primary alcohol is produced on reduction of an aldehyde.

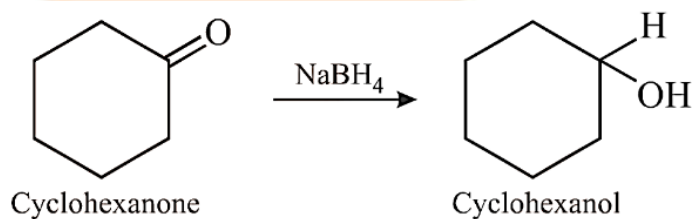
Example:



Reaction showing Catalytic Hydrogenation

2. Secondary alcohol is produced on reduction of a ketone.

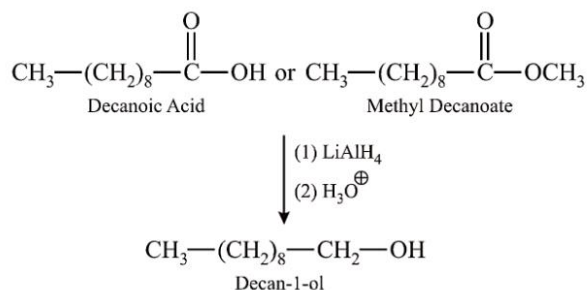
Example:



Reaction showing reduction of a ketone to form a Secondary alcohol.

3. Tertiary alcohol is produced on reduction of an acid or ester.

Example:



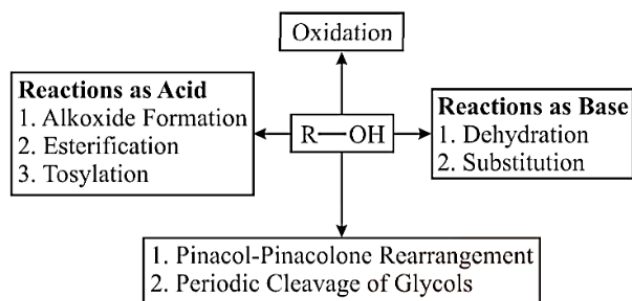
Reaction showing reduction of an acid or ester to form a Tertiary alcohol .

Reduction of LiAlH_4 and NaBH_4

Functional Group	Structure	NaBH_4	LiAlH_4
Aldehyde	$\text{R}-\overset{\text{O}}{\parallel}\text{C}-\text{H}$	$\text{R}-\text{CH}_2-\text{OH}$	$\text{R}-\text{CH}_2-\text{OH}$
Ketone	$\text{R}-\overset{\text{O}}{\parallel}\text{C}-\text{R}'$	$\text{R}-\underset{\text{OH}}{\text{CH}}-\text{R}'$	$\text{R}-\underset{\text{OH}}{\text{CH}}-\text{R}'$
Alkene	>C=C<	No Reaction	No Reaction
Acid Anion	$\text{R}-\overset{\text{O}}{\parallel}\text{C}-\text{O}^{\ominus}$ anion in base	No Reaction	$\text{R}-\text{CH}_2-\text{OH}$
Ester	$\text{R}-\overset{\text{O}}{\parallel}\text{C}-\text{OR}'$	No Reaction	$\text{R}-\text{CH}_2\text{OH}$ + $\text{R}'\text{OH}$

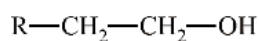
Reaction showing reduction of various functional groups in the presence

6. Reaction of Alcohols



Flow chart showing reactions of alcohols

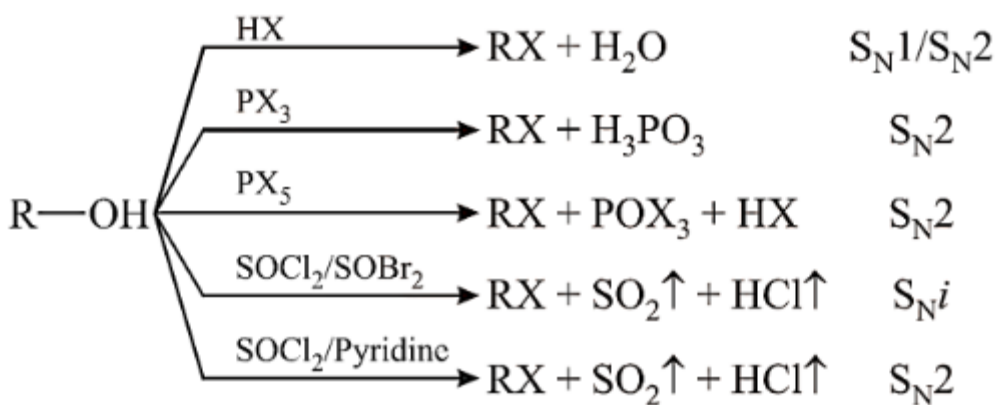
6.1 Dehydration



$\xrightarrow[\text{or conc. } H_3PO_4]{\text{conc. } H_2SO_4}$	Saytzeff alkene (With rearrangement)	E1/E2
$\xrightarrow[400^\circ-500^\circ C]{Al_2O_3}$	Saytzeff alkene (No rearrangement)	E2
$\xrightarrow{ThO_2}$	Hoffmann alkene (No rearrangement)	E2
$\xrightarrow{P_2O_5, \Delta}$	Saytzeff alkene (No rearrangement)	E2

Dehydration Reaction of alcohol

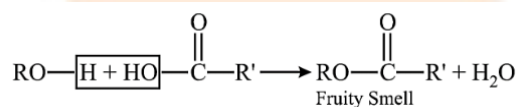
6.2 Substitution



Substitution Reaction of alcohol

6.3 Esterification

It is catalyzed by an acid or a base.

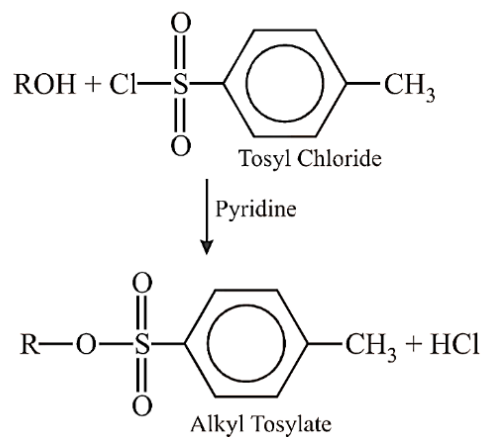


Esterification Reaction of alcohol

6.4 Tosylation

It is used to convert poor leaving group OH to good leaving group OTs.

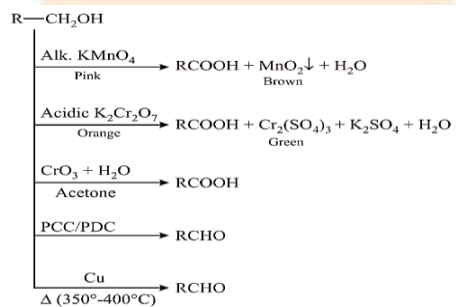
Example:



Tosylation Reaction of alcohol

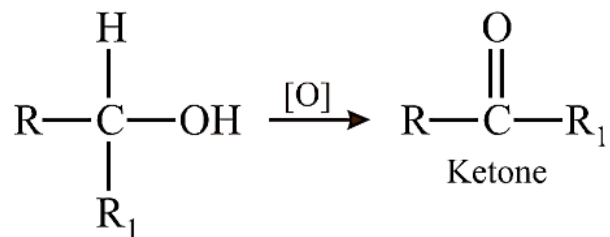
6.5 Oxidation

a. Primary Alcohols



Oxidation Reaction of primary alcohol

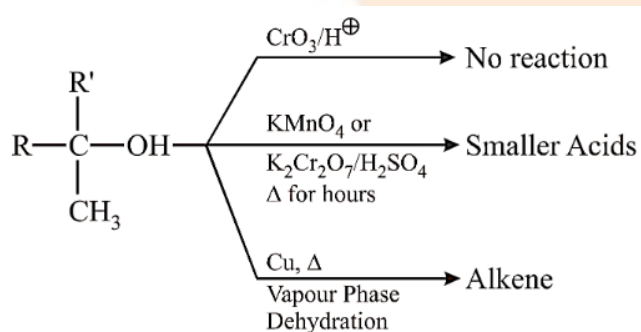
b. Secondary alcohols



Oxidation Reaction of Secondary alcohol

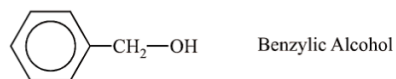
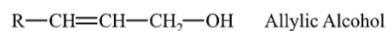
In the presence of any oxidizing agent, secondary alcohol is oxidized to alcohol.

c. Tertiary alcohols



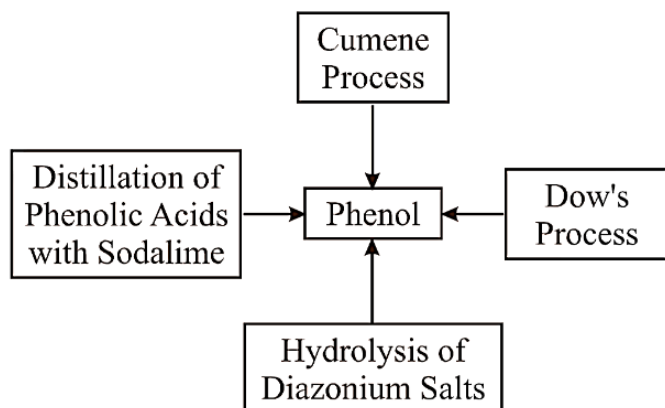
Oxidation Reaction of tertiary alcohol

MNO_2 is an oxidizing agent which is used to oxidise only benzylic, allylic and propargylic alcohols.



Showing benzylic, allylic and propargylic alcohols.

7. Preparation of Phenols

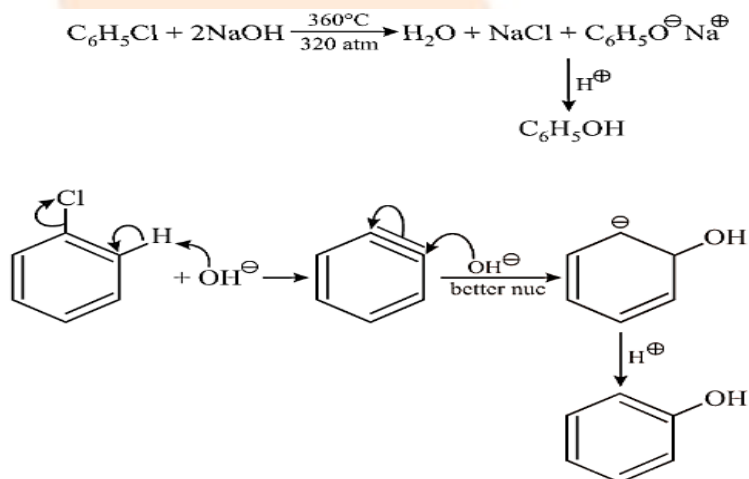


Flow chart showing various methods of preparation of phenol

7.1 Dow's Process

It is an industrial method which is used for the preparation of phenol. It takes place through a benzene mechanism.

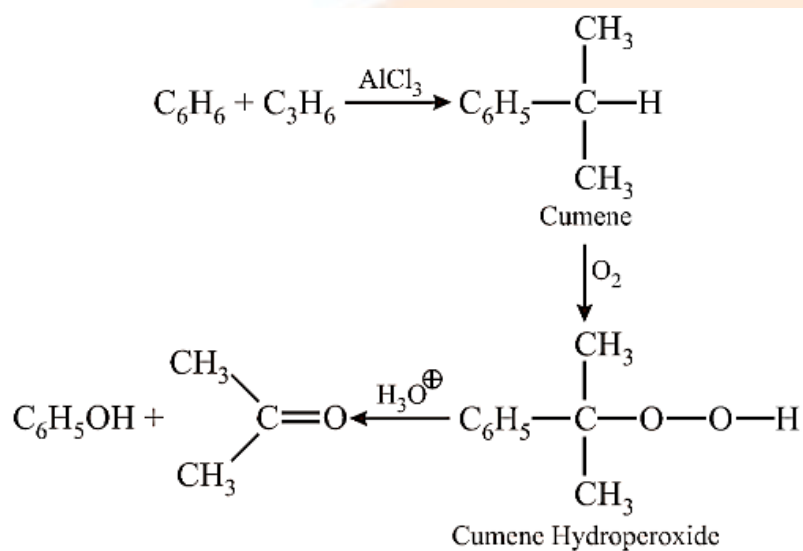
Example:



Showing Dow's Process

7.2 Cumene Process

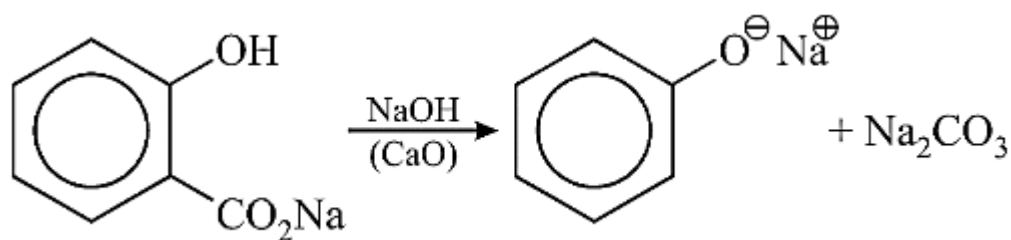
Example:



Showing Cumene Process

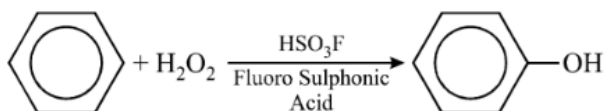
7.3 Distillation of phenolic acids with soda lime

Example:



7.4 Benzene

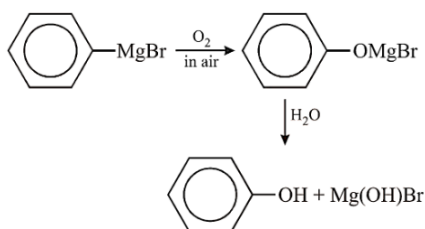
Example:



Showing reaction of Benzene with Hydrogen peroxide to yield phenol

7.5 Grignard Reagent

Example:

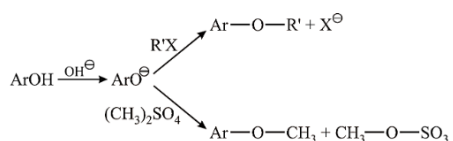


Showing the formation of phenol using grignard reagent

8. Reaction of Phenols

8.1 Formation of Ethers

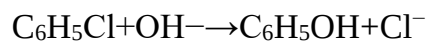
a. Williamson Synthesis



Showing the Reaction of phenols - Williamson synthesis

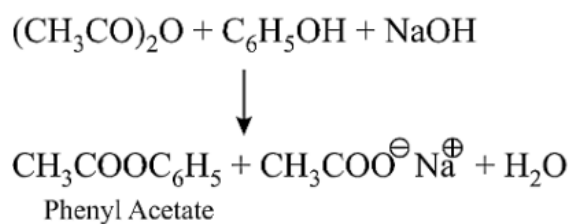
b. Nucleophilic Aromatic Substitution

Example:



8.2 Formation of Esters

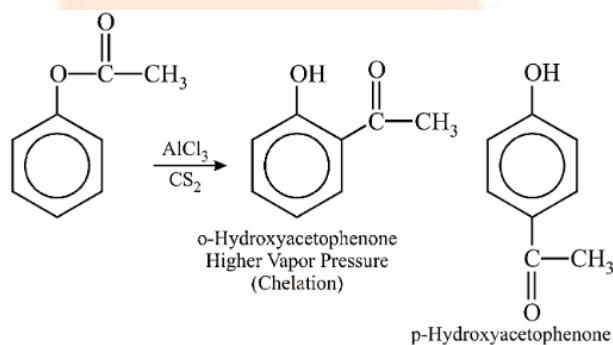
Example:



Showing the Reaction of phenols - Formation of Esters

8.3 Fries Rearrangement

Example:

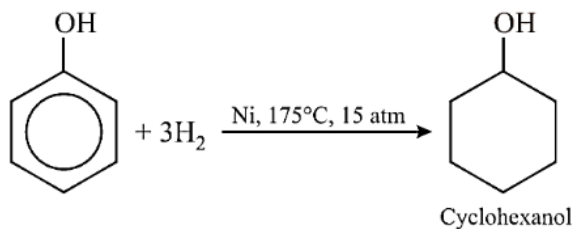


Showing the Reaction of phenols - Fries Rearrangement

8.4 Reactions of Benzene Ring

8.4.1 Hydrogenation

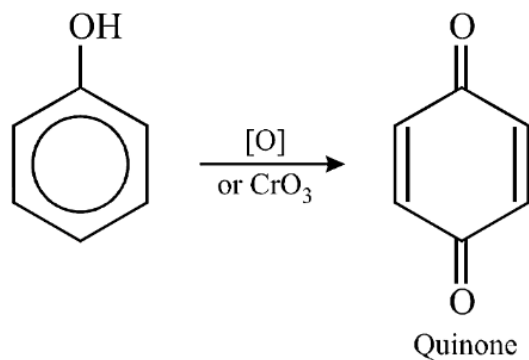
Example:



Showing the Reaction of phenols - Hydrogenation

8.4.2 Oxidation to Quinones

Example:

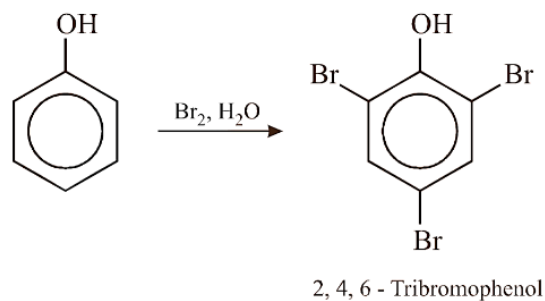


Showing the Reaction of phenols - Oxidation to quinones

8.4.3 Electrophilic Substitution

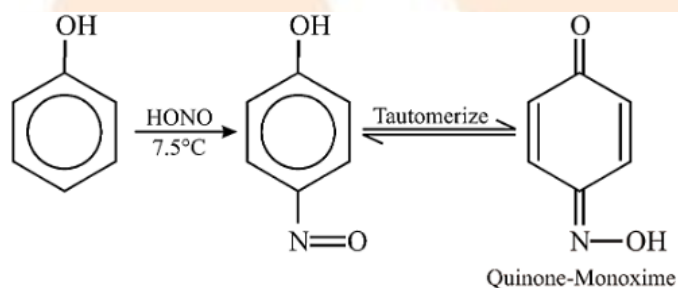
a. Halogenation

Example:



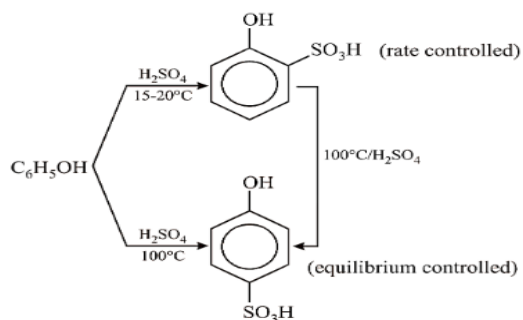
Showing the Reaction of phenols - Electrophilic substitution (a) Halogenation

b. Nitration



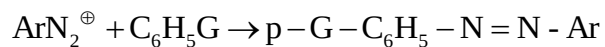
Showing the Reaction of phenols - Electrophilic substitution (b) Nitration

c. Sulphonation



Showing the Reaction of phenols - Electrophilic substitution (c) Sulphonation

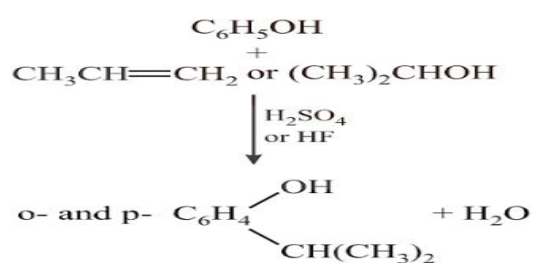
d. Diazonium Salt Coupling-Azophenols



Here, G is an electron releasing groups such as -OH, -OR.

e. Ring Alkylation

Example:

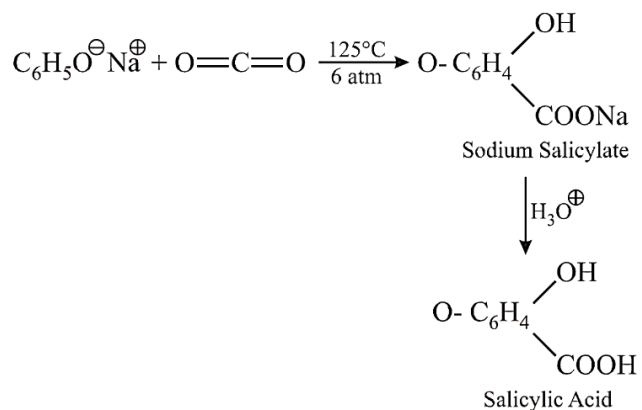


Showing the Reaction of phenols - Electrophilic substitution (e) Ring Alkylation

RX and AlCl_3 give poor yields as aluminum chloride coordinates with lone pairs of oxygen.

f. Kolbe's Synthesis

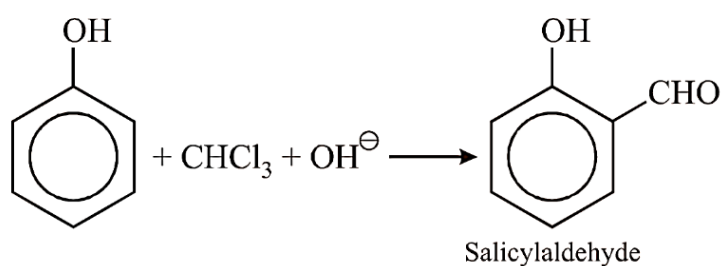
Example:



Showing the Reaction of phenols - Electrophilic substitution (f) Kolbe's Synthesis

g. Reimer-Tiemann Synthesis of phenolic Aldehydes

Example:



Showing the Reaction of phenols - Electrophilic substitution (g) Reimer-Tiemann Synthesis of phenolic Aldehydes

This reaction involves the formation of CCl_2 .

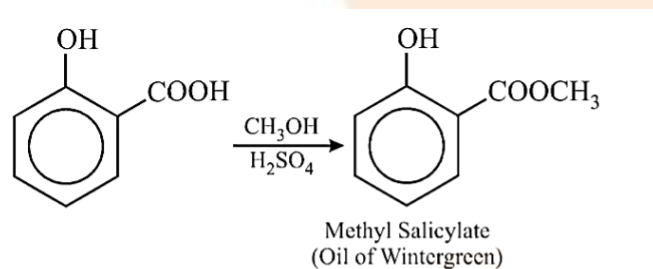
h. Formation of Aspirin

Example:

Acetic Anhydride+Salicylic Acid→Aspirin+Acetic Acid

Formation of oil of wintergreen

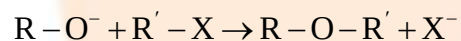
Example:



Showing the Reaction of phenols - Electrophilic substitution (j) Formation of oil of wintergreen

9. Ethers

9.1 Williamson Ether Synthesis

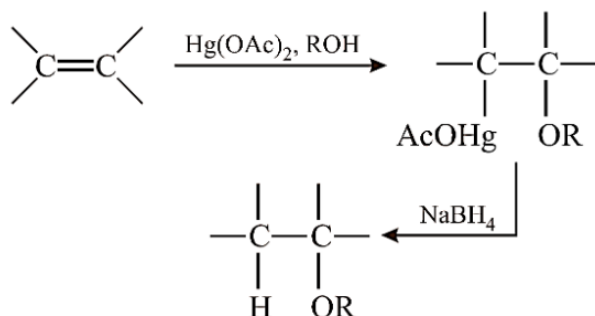


Note:

Substrate and leaving Group in Williamson synthesis

1. Leaving group, that is, $X = Cl, Br, I, OTs$ etc.
2. Substrate-Alkyl group R' should be primary.

9.2 Alkoxymercuration-Demercuration

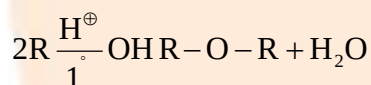


Showing the Alkoxymercuration-Demercuration

This type of reaction follows Markovnikov orientation.

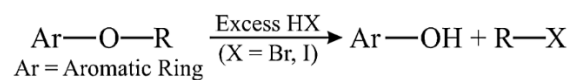
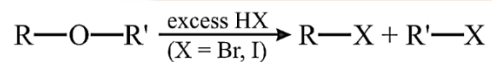
9.3 Bimolecular Dehydration of Alcohols

This is an industrial method which is used for the synthesis of ethers.



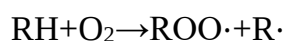
10. Reactions of Ethers

10.1 Cleavage by HBr and HI



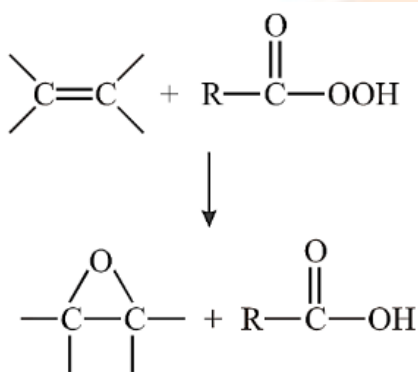
Showing the Cleavage by HBr and HI

10.2 Autoxidation



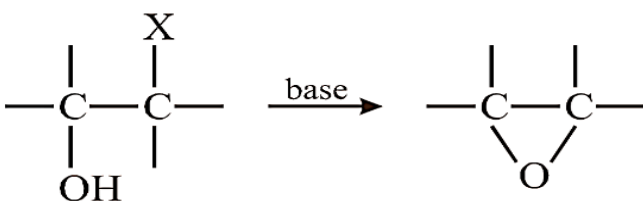
11. Preparation of Epoxide

11.1 Peroxy Acid Epoxidation



Showing the Preparation of epoxide-1 Peroxy Acid Epoxidation

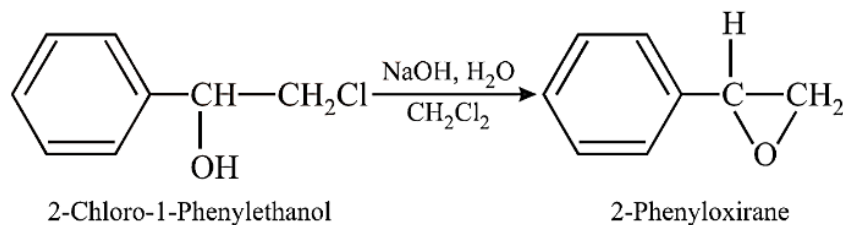
11.2 Base-Promoted Cyclization of Halohydrins



X = Cl, Br, I, OTs, etc.

Showing the Preparation of epoxide-2 Base-promoted cyclization of halohydrins

Example

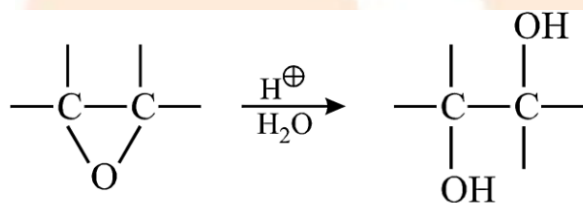


Showing the reaction of 2-Chloro-1-Phenylethanol to yield 2-Phenyloxirane

12. Reactions of Epoxides

12.1 Acid-Catalyzed Opening

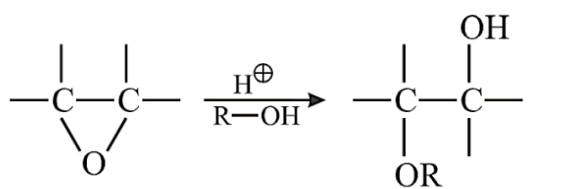
a. In Water



Showing the Reactions of Epoxides - (a) In water

Anti-stereochemistry is followed here.

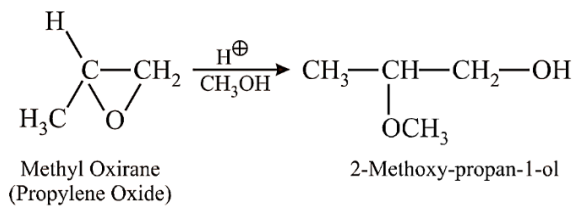
b. In Alcohols



Showing the Reactions of Epoxides - (b) In alcohols

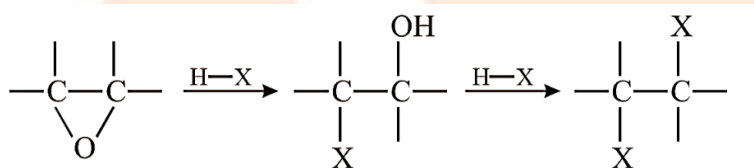
Here, the alkoxy group is bonded to the more highly substituted carbon.

Example:



Reaction of methyl Oxirane to yield 2-Methoxy-propan-1-ol

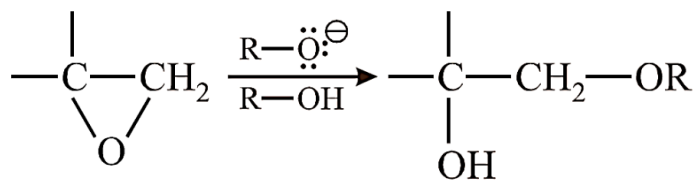
c. Hydrohalic Acids (X = Cl, Br, I)



Showing the Reactions of Epoxides - (c) Hydrohalic Acids (X = Cl, Br, I)

12.2 Base-Catalyzed Opening

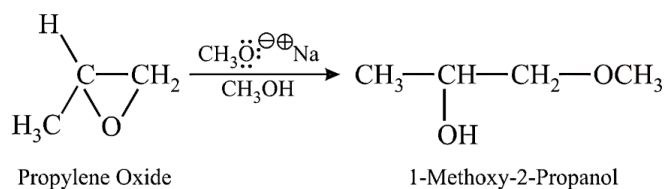
a. With Alkoxides



Showing the Reactions of Epoxides - 2 Base-Catalyzed opening (a) With Alkoxides

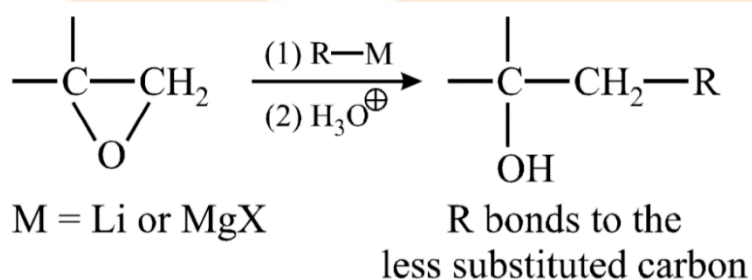
The alkoxy group is bonded to less highly substituted carbon.

Example:



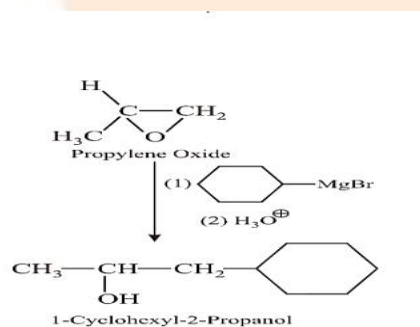
Reaction of Propylene oxide to yield 1-Methoxy-2-propanol

b. With Organometallics



Showing the Reactions of Epoxides - 2 Base-Catalyzed opening (b) With Organometallics

Example:



Reaction of Propylene oxide to yield 1-Cyclohexyl-2-Propanol

Note:

Opening of Epoxide Ring

1. In acid catalyzed opening, a nucleophile attacks the epoxide carbon, allowing a more stable carbocation to be formed.
2. Nucleophile attacks on the less hindered carbon in base catalyzed opening.

13. Acidic Strength

- I. Despite the fact that oxygen is more electronegative than sulfur, alcohols are weaker acids than thiols. RO^- , the conjugate base of alcohol, is more basic than RS^- because the negative charge in RO^- is placed on smaller oxygen atoms, resulting in higher charge density. However, because the RS^- negative charge is dispersed on larger sulphur, it is a poor base and its conjugate acid is more acidic.
- II. Due to the +I effect of the alkyl group, all alcohols (except CH_3OH) are weaker than H_2O . CH_3OH is slightly more powerful than H_2O . Because of the electron withdrawing benzene ring and the resonance stabilized phenolic ion, phenols are stronger than alcohol. Because alkoxide ions, the conjugate base of alcohol, lack resonance, they are less stable and more basic. Phenol is less stable than carboxylic acid, which has a strong electron-drawing carbonyl group and more stable, resonating structures.

14. Test for Alcohols, Phenols and Ethers

14.1 Analysis of Alcohols-Characterization

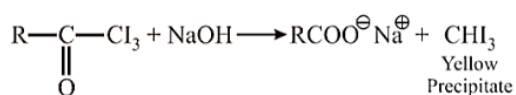
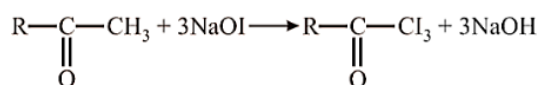
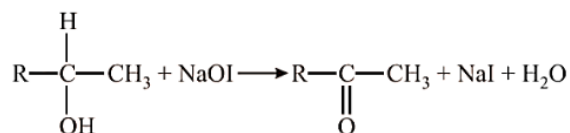
- a. Cold concentrated sulfuric acid dissolves alcohols. This property is shared by alkenes, amines, almost all oxygen-containing compounds, and easily sulfonated compounds. (Alcohol, like other oxygen-containing compounds, produces oxonium salts that dissolve in the highly polar sulfuric acid.)
- b. Cold dilute, neutral permanganate does not oxidize alcohols (although primary and secondary alcohols are oxidized by permanganate under more vigorous conditions). However, as we've seen, alcohols frequently contain impurities that oxidize under these conditions, so the permanganate test should be used with caution.
- c. Alcohols have no effect on the color of bromine in carbon tetrachloride. This characteristic distinguishes them from alkenes and alkynes.

- d. The evolution of hydrogen gas from alcohol reactions with sodium metal is useful in characterization.
- e. The formation of an ester upon treatment with an acid chloride or anhydride often indicates the presence of a hydroxide group in a molecule. Some esters have a pleasant odor; others have high melting points and can be used to make identifications.
- f. The Lucas test, which is based on the difference in reactivity of the three classes towards hydrogen halides, determines whether an alcohol is primary, secondary, or tertiary. Alcohols with fewer than six carbons are soluble in the Lucas reagent, which is a solution of concentrated hydrochloric acid and zinc chloride. The cloudiness that appears when the chloride separates from the solution indicates the formation of a chloride from an alcohol. As a result, the time required for cloudiness to appear is a measure of the alcohol's reactivity. The Lucas reagent reacts immediately with tertiary alcohol. Within five minutes, a secondary alcohol reacts. At room temperature, a primary alcohol does not react significantly. Benzyl alcohol and allyl alcohol react with the Lucas reagent as quickly as tertiary alcohols. Allyl chlorides, on the other hand, is soluble in the reagent.

Positive Iodoform Test	Negative Iodoform Test
$\begin{array}{c} \text{H} \\ \\ \text{CH}_3 - \text{C} - \text{H} \\ \\ \text{OH} \end{array}$	Any other primary alcohol
$\begin{array}{c} \text{H} \\ \\ \text{CH}_3 - \text{C} - \text{CH}_3 \\ \\ \text{OH} \end{array}$	$\begin{array}{c} \text{H} \\ \\ \text{CH}_3 - \text{C} - \text{CH}_3 \\ \\ \text{OH} \end{array}$
$\begin{array}{c} \text{H} \\ \\ \text{CH}_3 - \text{C} - \text{CH}_2\text{CH}_2\text{CH}_3 \\ \\ \text{OH} \end{array}$	$\begin{array}{c} \text{H} \\ \\ \text{CH}_3\text{CH}_2 - \text{C} - \text{CH}_2\text{CH}_3 \\ \\ \text{OH} \end{array}$
$\begin{array}{c} \text{H} \\ \\ \text{C}_6\text{H}_5 - \text{C} - \text{CH}_3 \\ \\ \text{OH} \end{array}$	$\text{C}_6\text{H}_5 - \text{CH}_2 - \text{CH}_2\text{OH}$

Showing which alcohol gives positive or negative Iodoform test.

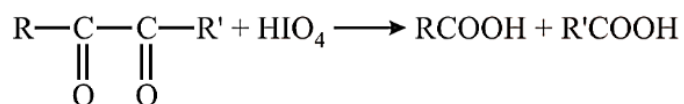
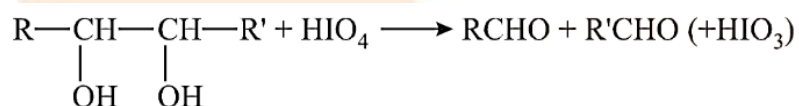
Oxidation, halogenation, and cleavage are all involved in the reaction.



Showing Oxidation, halogenation, and cleavage reactions.

14.2 Analysis of Glycols, Periodic Acid Oxidation

$\text{HI}\{\text{O}_4\}$ compounds with two or more -OH or C=O groups attached to adjacent carbon atoms undergo oxidation with cleavage of carbon-carbon bonds when exposed to periodic acid.

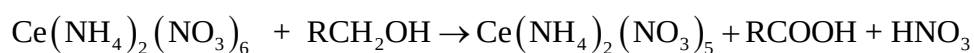


Showing Analysis of Glycols, Periodic Acid Oxidation

14.3 Miscellaneous Tests

a. Ceric Ammonium Nitrate Test-

With this reagent, alcohols produce a red color.



b. Potassium Dichromate Test –

Alcohols change the color of orange dichromate to green. Tertiary alcohols fail this test.

c. Ester test:

Alcohol produces a fruity aroma of ester with carboxylic acid.

d. Methanol reacts with salicylic acid to produce winter green oil.

14.4 Victor Meyer's Test

Victor Meyer's Test

Victor Meyer's test is used to distinguish between primary, secondary, and tertiary alcohols based on their reaction with a reagent to form colored complexes.

Reaction for Primary Alcohols:

1. Reaction with Chlorine (Cl_2) or Bromine (Br_2): $\text{RCH}_2\text{OH} \xrightarrow{\text{Cl}_2/\text{Br}_2} \text{RCHCl}_2 / \text{RCHBr}_2$

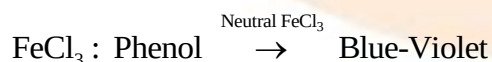
2. Reaction with Victor Meyer's Reagent (a mixture of iodine and potassium iodide in the presence of a strong base, e.g., sodium hydroxide): $(\text{RCH}_2\text{I} + \text{I}_2 + \text{KOH} \rightarrow \text{RCH}_2\text{I}_2 + \text{KI})$

14.5 Differentiation Test

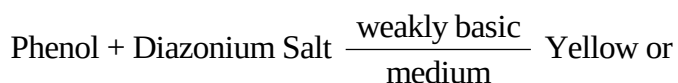
14.5.1 Alcohols and Phenols

1. **Litmus Test:** Phenol turns blue litmus red but not alcohol.

2. **Ferric Chloride:**

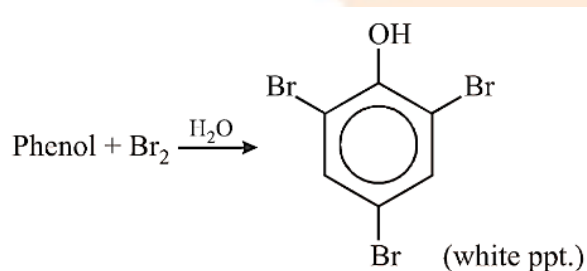


3. **Coupling Reaction:**



Diazonium Salt + Alcohol $\xrightarrow[\text{medium}]{\text{weakly basic}}$ No Reaction

4. Bromine Water Test:



Alcohol + Br₂ $\xrightarrow{\text{H}_2\text{O}}$ No Reaction

Showing Bromine Water test

14.4.2 Alcohols and Ethers

- Alcohols react with Na to give H₂, but not ethers.
- Alcohol gives fumes of HCl with PCl₅ but not ethers.

14.5.3 Sodium Bicarbonate Test

Phenol, ROH and H₂O do not displace CO₂ from carbonate & bicarbonates but RCOOH & RSO₃H gives brisk effervescence of CO₂ which proves that RCOOH & RSO₃H are stronger acids than H₂CO₃ but phenol is weaker acid than H₂CO₃. Nitrophenols also give effervescence of CO₂ with Na₂CO₃. Trinitrophenol (Picric Acid) is highly acidic due to strong electron withdrawing effect of three groups its acidic strength is comparable to that of carboxylic acids. Its anion is highly resonance stabilised.



1. For Carboxylic Acids:



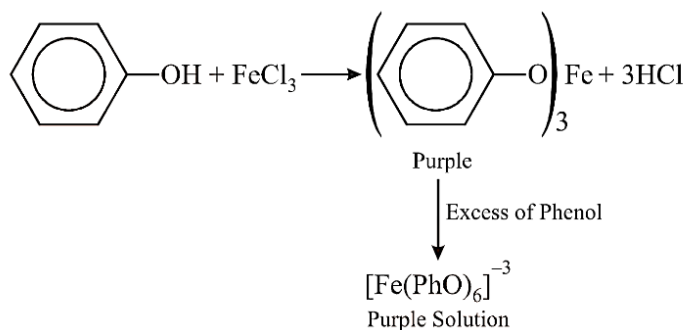
Observation: Effervescence due to the release of carbon dioxide (CO_2) gas indicates the presence of a carboxylic acid.

2. For Alcohols and Phenols:

Alcohols and phenols do not react with sodium bicarbonate, so no effervescence is observed.

15.5.4 FeCl_3 Test:

Phenol gives characteristic purple colour with FeCl_3 but alcohols do not react with FeCl_3 . Carboxylic acids also form buff coloured precipitate with FeCl_3 . Only acetic acid forms red coloured precipitate with FeCl_3 , so it can be used as a test for acetate salts.



Showing Ferric Chloride Test in case of phenols